

UKIAH, CA, USA

WMO: 725905

Lat: 39.126N

Lon: 123.201W

Elev: 601

StdP: 14.38

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23275

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.5	30.6	21.1	16.2	49.2	24.8	19.3	46.4	18.1	51.6	16.1	50.9	0.3	310	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.6	100.4	67.4	97.4	66.6	93.9	65.6	70.2	95.2	68.5	92.8	67.1	90.7	7.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	81.2	79.5	58.7	75.4	77.2	57.1	71.1	75.1	34.3	95.3	32.9	93.0	31.7	90.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.4	12.1	DB	24.8	106.7	3.1	2.6	22.6	108.6	20.7	110.1	19.0	111.5	16.7	113.4	(4)
(5)			WB	23.7	73.2	3.1	1.9	21.5	74.6	19.7	75.8	18.0	76.9	15.7	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	47.5	49.3	52.7	55.8	62.3	69.0	74.8	73.6	69.6	61.1	51.8	46.3	(6)	
	DBStd	11.36	4.70	5.27	5.89	6.40	6.26	6.38	5.12	4.54	5.62	5.57	5.58	5.07	(7)	
	HDD50	410	104	70	38	16	1	0	0	0	0	1	44	136	(8)	
	HDD65	2950	542	439	381	284	130	30	1	1	20	148	395	579	(9)	
	CDD50	3893	27	51	123	189	381	569	767	732	588	344	99	23	(10)	
	CDD65	957	0	0	1	7	45	149	303	268	158	26	0	0	(11)	
	CDH74	14783	0	7	97	311	961	2335	4218	3703	2408	711	32	0	(12)	
	CDH80	7839	0	0	15	96	404	1194	2456	2102	1299	271	2	0	(13)	
(14) Wind	WSAvg	3.7	2.6	3.4	3.9	4.5	4.8	5.1	4.6	4.0	3.4	2.7	2.3	2.6	(14)	
(15) Precipitation	PrecAvg	42.0	8.3	6.8	5.6	2.6	1.0	0.3	0.0	0.2	0.5	2.0	5.9	8.0	(15)	
	PrecMax	82.4	25.0	22.3	18.7	9.8	6.4	2.2	0.9	1.5	3.2	8.2	17.4	22.4	(16)	
	PrecMin	15.0	0.5	0.4	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	(17)	
	PrecStd	12.8	5.6	5.0	4.3	2.2	1.4	0.5	0.2	0.4	0.8	1.9	4.3	6.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.9	74.5	82.0	88.2	95.5	102.4	105.5	104.3	101.5	93.4	78.9	65.1	(19)	
		MCWB	53.3	56.7	59.6	61.4	65.6	69.1	69.5	67.7	65.2	62.4	58.6	52.3	(20)	
	2%	DB	63.8	69.2	76.2	82.4	89.7	97.1	100.4	99.3	97.1	87.0	72.5	61.0	(21)	
		MCWB	51.8	54.4	56.9	60.1	63.3	67.3	68.1	66.8	64.8	60.5	56.3	52.3	(22)	
	5%	DB	59.4	64.0	71.5	76.9	84.3	91.7	97.1	95.4	92.8	82.0	66.5	57.9	(23)	
		MCWB	51.2	51.9	55.3	58.2	61.6	65.1	67.5	65.5	63.9	59.1	54.0	52.0	(24)	
	10%	DB	55.9	59.2	65.8	70.8	79.2	87.2	93.0	91.6	88.4	76.9	62.0	55.5	(25)	
		MCWB	50.8	51.0	52.9	55.3	59.6	63.5	66.5	65.1	62.9	57.5	53.0	50.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	59.1	61.5	63.9	67.9	72.2	72.9	70.6	69.5	65.2	61.3	57.8	(27)	
		MCDB	62.1	69.4	78.9	83.7	91.1	97.6	98.7	97.4	95.6	84.9	74.0	60.3	(28)	
	2%	WB	54.8	56.3	58.7	60.7	65.2	69.0	70.7	68.6	67.1	62.8	58.4	55.8	(29)	
		MCDB	58.5	64.7	73.7	79.4	86.5	92.9	95.7	94.0	91.2	81.2	67.4	58.1	(30)	
	5%	WB	53.3	54.1	56.2	58.8	62.7	66.6	69.0	67.4	65.2	60.8	56.3	53.8	(31)	
		MCDB	56.9	60.7	67.7	75.3	82.0	88.6	93.3	92.1	89.1	77.8	63.2	56.1	(32)	
	10%	WB	51.4	52.1	54.0	56.3	60.2	64.4	67.4	66.0	63.4	58.9	54.4	51.7	(33)	
		MCDB	55.4	57.8	63.0	69.3	76.9	84.7	90.9	89.6	85.8	73.4	60.1	54.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	20.0	22.7	24.7	27.9	31.2	35.6	36.4	35.4	30.7	21.6	18.3	(35)	
		MCDBR	27.4	30.8	34.8	36.9	38.9	39.8	40.9	41.7	43.3	41.0	31.5	23.9	(36)	
	5% WB	MCWBR	16.5	17.0	17.2	17.1	16.2	15.2	15.0	15.1	16.4	17.3	16.6	15.1	(37)	
		MCWBR	18.6	24.4	29.6	33.5	35.3	37.0	38.0	38.5	39.0	34.9	24.3	14.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.316	0.319	0.324	0.346	0.339	0.343	0.341	0.332	0.328	0.320	0.312	0.317	(40)	
		taud	2.559	2.550	2.548	2.445	2.505	2.501	2.508	2.525	2.525	2.588	2.582	2.552	(41)	
	Ebn at Noon	Ebn at Noon	274	289	298	295	298	295	295	296	292	285	273	264	(42)	
		Edh at Noon	23	27	30	35	34	34	33	32	30	25	23	22	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	692	980	1393	1864	2245	2501	2422	2133	1765	1250	816	589	(44)	
	RadStd	RadStd	112	138	145	160	141	120	87	81	75	95	73	104	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.70	N/A	-1.94	N/A	-0.68	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	-2.43	N/A	N/A	N/A	N/A	N/A	+243	+70

Nomenclature: See separate page